

SYNOPSIS

At 1658 on 29 August 2015, the Cyprus registered cargo ship *Daroja* and the St Kitts and Nevis registered oil bunker barge *Erin Wood* collided 4 nautical miles south-east of Peterhead, Scotland. Minor damage was caused to *Daroja* but damage to *Erin Wood* included breaches of the hull, resulting in flooding of the vessel and pollution from leaking fuel cargo.

At the time of the accident, both vessels were manned by watchkeepers not keeping a lookout and therefore unaware of the risk of collision. On board *Daroja*, the chief officer, who was the officer of the watch, missed opportunities to detect *Erin Wood* by visual, radar and automatic identification system means. This happened because he had become complacent through distraction, inattentiveness, the repetitive nature of the vessel's schedule and insufficient supervision. On board *Erin Wood*, the bridge was not continuously manned and, although the skipper was aware of the presence of another vessel, the situation was not effectively assessed and an assumption was made that a larger ship would keep clear.

Lone watchkeeping was a causal factor on board both vessels and, similar to previous MAIB investigations, this report highlights the importance of identifying and managing risks when a sole watchkeeper is the only lookout.

This investigation has also identified significant safety shortcomings in the management and operation of *Erin Wood*. The vessel's crew did not have the competence necessary to operate a small tanker and there was not an effective safety management system. This unsafe situation arose because *Erin Wood*'s managing company lacked experience in the industry sector and prioritised commercial gain ahead of safety at sea. Risks associated with *Erin Wood*'s operations also went undetected by the flag and coastal states. This happened because the Flag State's process of initial registration was not sufficiently thorough and because *Erin Wood* was not inspected under the Port State Control regime.

United Shipping Companies Barnkrug GmbH & Co.KG, *Daroja*'s managing company, has reviewed its safety management system and made a series of changes to improve bridge watchkeeping standards. Northern Oils Limited, *Erin Wood*'s managing company, has temporarily ceased transportation of fuel products by sea. Nevertheless, safety recommendations have been made to both managing companies intended to improve standards of navigation and watchkeeping. A safety recommendation has also been made to the St Kitts and Nevis International Ship Registry intended to improve its assessment of potential risks during the process of initial registration of vessels being taken onto its flag.

SECTION 1 – FACTUAL INFORMATION

1.1 PARTICULARS OF *DAROJA*, *ERIN WOOD* AND THE ACCIDENT

SHIP PARTICULARS		
Vessel's name	<i>Daroja</i>	<i>Erin Wood</i>
Flag	Republic of Cyprus	St Kitts and Nevis
Flag State's Recognised Organisation	Det Norske Veritas-Germanischer Lloyd	International Register of Shipping
IMO number	9148221	Not IMO registered
Type	Cargo ship	Oil bunker barge
Registered owner	Marjesco Schiffsbeteiligungs GmbH	Northern Oils Limited
Manager(s)	United Shipping Companies Barnkrug GmbH & Co.KG	Northern Oils Limited
Construction	Steel	Steel
Year of build	1997	1964
Registered Length	85.0m	25.3m
Gross tonnage	3266	70
Minimum safe manning	7	2
Authorised cargo	Containers	Marine gas oil
VOYAGE PARTICULARS		
Port of departure	Aberdeen, Scotland	North Shields, England
Intended port of arrival	Lerwick, Scotland	Scrabster, Scotland
Type of voyage	Short domestic	Short domestic
Cargo information	181 containers of general cargo	130m ³ of marine gas oil
Manning	9	2
MARINE CASUALTY INFORMATION		
Date and time	1658 on 29 August 2015	
Type of marine casualty	Very Serious Marine Casualty	
Location of incident	57°26.36'N – 001°41.58'W	
Place on board	Hull Bulbous bow	Hull, engine room, accommodation and superstructure
Injuries/fatalities	None	None
Damage/environmental impact	Indentation and scraping to bulbous bow	Shell plating fractures, flooding, superstructure damage, fuel cargo pollution
Ship operation	On passage	On passage
Voyage segment	Mid water	Mid water
External & internal environment	Wind: South-westerly, 4 to 6 knots, sea state 2, 0.5m swell, visibility good	
Persons on board	9	2

1.2 NARRATIVE

1.2.1 Events prior to the collision

At 2130 on 28 August 2015, the oil bunker barge *Erin Wood* sailed from North Shields, England and commenced passage to Scrabster, Scotland (**Figure 1**). The vessel was loaded with 130m³ of marine gas oil and was manned by its skipper and a deckhand. At 0100 (29 August 2015), with the vessel proceeding on autopilot at a passage speed of 9.5kt, the skipper handed the bridge watch over to the deckhand and went to his cabin to rest. The skipper returned to the bridge at about 0830 and took back control of the watch; the deckhand then left the bridge and went to rest.

At 1510 on 29 August 2015, the general cargo vessel *Daroja* sailed from Aberdeen, Scotland and commenced passage to Lerwick in the Shetland Islands (**Figure 1**). Once clear of Aberdeen Harbour, with the passage speed set at 14.5kt, *Daroja's* master handed the watch over to the second officer, then left the bridge. At 1538, the second officer altered *Daroja's* autopilot controlled heading to 034° (**Figure 2**). At 1600, *Daroja's* chief officer arrived on the bridge and, following a short handover brief from the second officer, took over the watch. Once alone on the bridge, the chief officer checked the bridge equipment settings and looked out of the windows to scan the horizon. At about 1620, the chief officer went to the chart table on the starboard side of the bridge and began working on paperwork relating to the cargo loaded in Aberdeen.

At about 1620, *Erin Wood's* skipper observed several automatic identification system (AIS) shipping tracks on the vessel's electronic chart system (ECS) display. The skipper interrogated the tracks to establish the closest point of approach (CPA) for each of them, and noted that the nearest CPA was predicted to be 1 nautical mile (nm). At 1625, *Erin Wood's* skipper adjusted his autopilot controlled heading to 350° (**Figure 2**); he then left the bridge and went to the stern deck to urinate into the sea. While on deck, the skipper noticed a larger vessel approaching from astern. He assumed the vessel would keep clear and made no attempt to assess it further. The skipper then collected his tablet computer from his cabin and returned to the bridge. About 10 minutes later, he went to the mess room and briefed the deckhand on the intended passage plan.

At 1646, the chief officer on board *Daroja* answered a call on the bridge internal telephone. Following a 4-minute discussion, the chief officer went to the starboard side of the bridge and sat in the bridge chair (**Figure 3**). At about the same time, *Erin Wood's* skipper returned to the bridge and sat down in the chair on the port side (**Figure 4**).

1.2.2 The collision

At 1658, *Daroja's* bulbous bow struck *Erin Wood's* port side (**Figure 5**). The bunker barge became lodged under *Daroja's* bow and started being driven sideways; in an attempt to break his vessel free, *Erin Wood's* skipper immediately deselected the autopilot, applied maximum port rudder and put the engine to full ahead. Within seconds, *Erin Wood* heeled over 90° to starboard and seawater rushed into its bridge, accommodation areas and engine room through the vessel's open weathertight doors. The skipper escaped from the flooded bridge through an open window; meanwhile, the deckhand, who was in the mess room, was fully submerged in seawater.

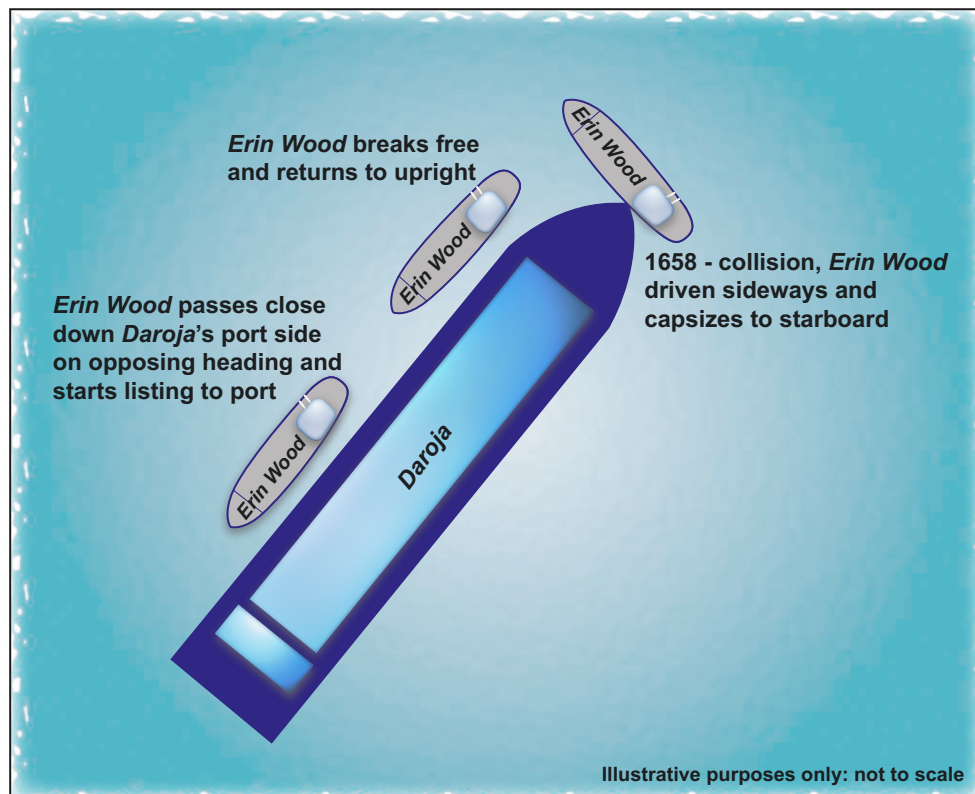


Figure 5: Schematic diagram of the collision sequence

After about 15 seconds, *Erin Wood* broke free, rolled back upright and passed down *Daroja*'s port side on an opposing heading (**Figure 5**). As *Erin Wood* came upright, the skipper found himself clinging to the bridge roof. The deckhand was washed out of the mess room and over the ship's side as the floodwater rushed back out through the open door. The deckhand grabbed hold of the top edge of the bulwark to prevent himself being swept completely overboard. When the rush subsided, the deckhand was able to climb back over the bulwark onto the vessel's upper deck.

1.2.3 Actions following the collision

When the collision happened, *Daroja*'s chief officer heard an unusual noise and stood up from the bridge chair and looked out ahead and to starboard. With nothing in sight, he then went to the port bridge wing, where he saw *Erin Wood* passing close down the port side. The master, who was in his cabin, had also heard the noise and phoned the bridge to find out what it was. The chief officer answered the phone and asked the master to come to the bridge immediately. The master arrived on the bridge at 1701 and called *Erin Wood* on very high frequency (VHF) radio channel 16, but there was no response. *Daroja*'s master also reduced the vessel's speed, turned back towards *Erin Wood* and ordered the fast rescue boat (FRB) to be prepared for launch.

Once clear of *Daroja*, *Erin Wood* settled low in the water and started listing to port as seawater flooded into No.4 port water ballast tank (WBT). The skipper, concerned that the vessel might sink or capsize, released the liferaft from the bridge roof and threw it over the side. He then climbed down to the deck and, with the assistance of the deckhand, inflated the liferaft and secured its painter to a cleat on the starboard quarter ready for immediate use. The deckhand then fetched two lifejackets from the accommodation space and both he and the skipper donned them.

It was apparent to *Erin Wood*'s skipper that the vessel was taking on water so, at 1704, he called Peterhead Harbour vessel traffic service (VTS) using VHF radio and stated that his vessel was sinking and help was urgently required. Peterhead VTS immediately relayed this information to Aberdeen Coastguard, who ordered the launch of the Peterhead RNLI¹ lifeboat and also tasked the oil rig support vessel *Grampian Talisman*, which was nearby, to proceed to the scene.

Daroja's FRB was launched at 1710 and proceeded directly to *Erin Wood*. The skipper of *Erin Wood* told the FRB coxswain that there were two crew on his vessel, both were safe and that no assistance was required. Shortly thereafter, *Erin Wood*'s skipper informed the coastguard that he assessed the vessel was no longer in immediate danger but that its engine room was partially flooded. *Erin Wood*'s crew then used an emergency salvage pump to start pumping floodwater out of the engine room; this was initially successful but the suction soon became blocked by rags.

The Peterhead RNLI lifeboat arrived at 1732 (**Figure 6**) and transferred two crewmen and a salvage pump onto *Erin Wood*. The RNLI salvage pump was used to assist the crew of *Erin Wood* in their efforts to pump out the floodwater from the

Image courtesy of the RNLI



Figure 6: *Erin Wood* after the collision as seen from the Peterhead lifeboat, including evidence of fuel cargo pollution

¹ Royal National Lifeboat Institution

engine room and accommodation space. Once the majority of the floodwater had been pumped out of the engine room, *Erin Wood*'s skipper turned his attention to addressing the vessel's list to port. This was corrected by pumping 11t of seawater into No.4 starboard WBT: 8t from the after peak tank and 3t from the sea (**Figure 7**).

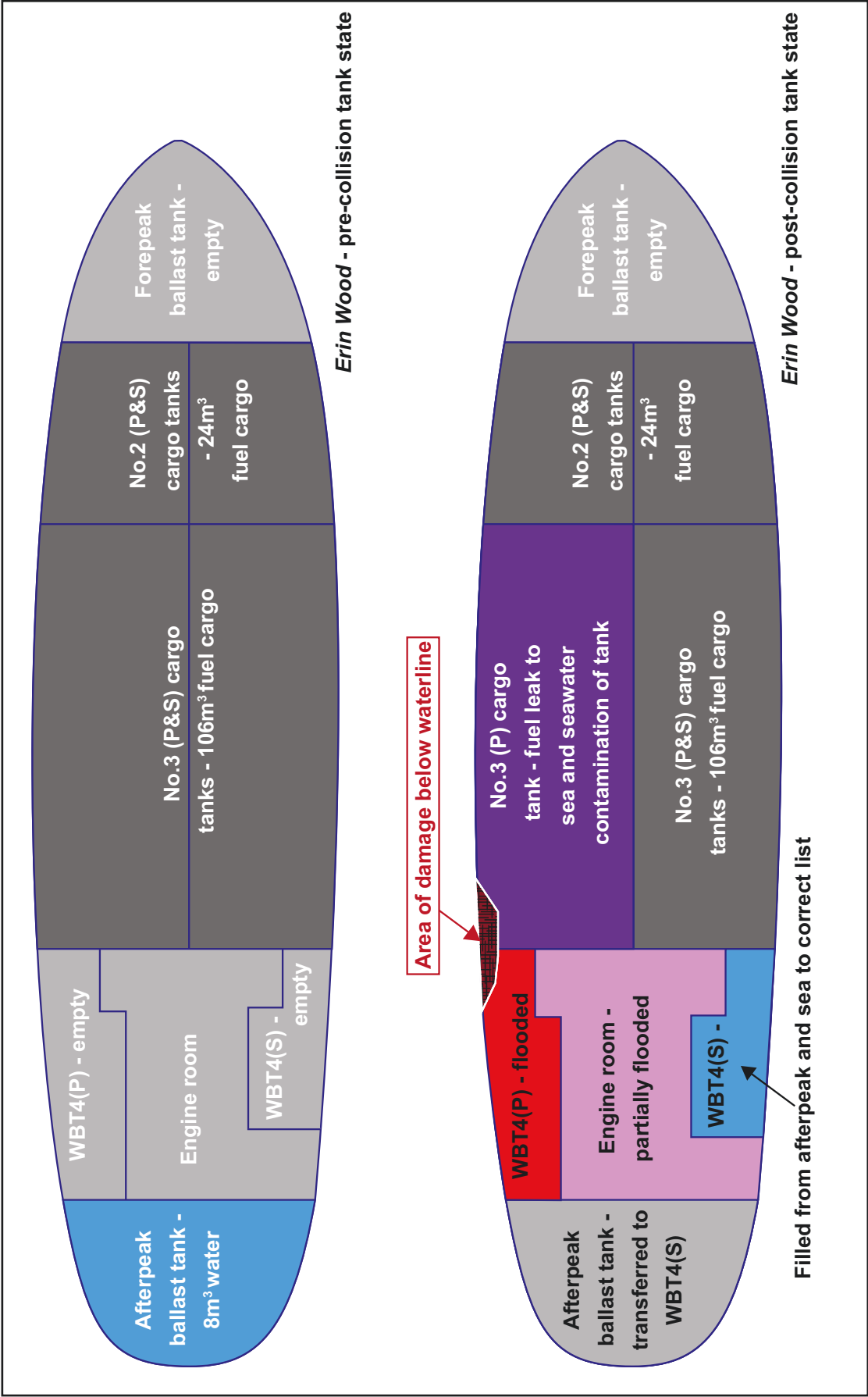


Figure 7: *Erin Wood* – general arrangements of tanks illustrating contents before and after collision

At 1830, Aberdeen Coastguard released *Daroja* from the scene but directed it to proceed alongside in Peterhead, where it arrived at 1945. The fishing vessel *Ocean Endeavour* arrived on scene at 1842 and started preparing to take *Erin Wood* in tow. *Grampian Talisman* was stood down by the lifeboat coxswain at 1941. Escorted by the lifeboat and under tow by *Ocean Endeavour* (**Figure 8**), *Erin Wood* berthed in Peterhead at about 2230.

Image courtesy of the RNLI



Figure 8: Fishing vessel *Ocean Endeavour* with *Erin Wood* in tow, taken from the Peterhead lifeboat

After arriving in Peterhead, both *Daroja* and *Erin Wood* were attended by local Police. The master and chief officer of *Daroja* and both crewmen from *Erin Wood* were breathalysed, all with negative results. An environmental containment boom was rigged around *Erin Wood* to prevent the spread of fuel leaking from the vessel.

1.3 ENVIRONMENTAL CONDITIONS

The environmental conditions in the vicinity of the accident were:

- Wind: south-westerly, 4 to 6kt
- Weather: overcast
- Sea state: smooth with approximate swell height of 0.5m
- Visibility: good

SECTION 2 – ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 THE COLLISION

Both *Daroja*'s chief officer and the skipper of *Erin Wood* had obligations under the COLREGs to take action to avoid collision. *Daroja* was the give-way vessel in a crossing situation (**Figure 22**) and the most appropriate action would have been for its course to have been altered to starboard to pass safely around *Erin Wood*'s stern.

Erin Wood's skipper, as the OOW on the stand-on vessel, had an initial obligation to hold course and speed. Had *Erin Wood*'s skipper been monitoring the situation, it would have become apparent that *Daroja*, as the give-way vessel, was not taking appropriate avoiding action. In this situation, *Erin Wood*'s skipper was obliged to act to avoid collision. However, because neither vessel was keeping a lookout, the risk of collision went undetected, no avoiding action was taken and the accident resulted.

2.3 THE FLOODING OF *ERIN WOOD*

When the collision happened, *Erin Wood*'s hull plating was damaged and the vessel heeled heavily to starboard. The collision damage caused the previously empty No.4 port WBT to free-flood and marine gas oil to leak from No.3 port cargo tank. *Erin Wood*'s upper deck weathertight doors had been left open and therefore, when the vessel heeled over, water rushed into the accommodation, bridge and engine room spaces.

The extent of the flooding placed the vessel and the lives of both crewmen in grave danger. It is likely that the instinctive reactions of the skipper to escape through a bridge window and the crewman to grab the top edge of the bulwark as he was washed out of the mess room prevented loss of life.

Erin Wood's external doors were routinely left open at sea and there was no procedure or guidance on board for the maintenance of weathertight integrity. However, the importance of ensuring the weathertight integrity of a low freeboard vessel such as *Erin Wood* should have been readily apparent to the crew. Had these doors been shut at the time of collision, the downflooding of the bridge, accommodation and engine room spaces would have been avoided.

2.4 BRIDGE ROUTINES AND LOOKOUT PROCEDURES ON BOARD *DAROJA*

Daroja's chief officer had been on the bridge as the OOW for almost an hour prior to the collision and was sitting in the bridge chair when the vessels collided (**Figure 3**). Other than scanning the horizon when he first came on watch, at no point thereafter did he keep a visual lookout. In addition, VDR evidence indicated that he did not interact with the radar display during that time.

SECTION 3 – CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. *Daroja* and *Erin Wood* collided because a proper lookout was not being kept on either vessel. [2.2]
2. On board *Daroja*, the chief officer, who was the OOW, missed multiple opportunities to detect *Erin Wood*; this happened because he had become complacent about his watchkeeping duties and allowed himself to become distracted. [2.4]
3. Complacency and poor watchkeeping practices were systemic on board *Daroja*. This was largely due to the repetitive nature of its trading route and a lack of mentorship and direction from the vessel's master. [2.4]
4. Although *Erin Wood*'s skipper was aware of the presence of another vessel, he did not effectively assess the situation and assumed a larger vessel would keep clear. [2.5]
5. Lone watchkeeping was a normal practice in both vessels and the risks associated with this had not been properly assessed. [2.6]
6. The lives of *Erin Wood*'s crew were placed in significant danger. The skipper's presence of mind to escape from the flooded bridge, and the deckhand managing to hold on to the bulwark to prevent being washed completely overboard, are actions that probably saved their lives. [2.3]
7. The flooding of *Erin Wood*'s bridge, accommodation spaces and engine room would have been prevented if the upper deck weathertight doors had been shut. [2.3]
8. *Erin Wood*'s crew did not have the competence necessary to operate a small coastal tanker; the vessel was also not provided with an effective safety management system. [2.8]

3.2 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. Northern Oils, the managers of *Erin Wood*, did not have the management experience necessary to run a fleet of small tankers. Additionally, the company prioritised commercial gain ahead of marine safety. [2.9.1]
2. Northern Oils did not recognise or take into account the risks associated with a grey listed Flag State and a low performing recognised organisation. [2.9.2]
3. Surveys of *Erin Wood*, conducted by the International Register of Shipping on behalf of the Flag State, did not ensure that the vessel complied with appropriate international, or applicable national, regulations. [2.10.1]

4. The St Kitts and Nevis International Ship Registry did not identify risks associated with *Erin Wood*'s operations due to inconsistencies in the initial survey regime and the absence of a dedicated Flag State inspection. [2.10.1]
5. The Maritime and Coastguard Agency did not identify the risks associated with *Erin Wood*'s operations primarily because the vessel was not IMO registered and, therefore, not inspected under the Port State Control regime. [2.10.2]